

Project-III

EE619

03/Apr/2026

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- This is a group project. Please do the project in the same team which executed project-1 and project-2.
 - This project involves the design, layout and characterization of two commonly used logic gates.
 - Please divide the work between the designers, such that the roles/responsibilities of the two members can be clearly identified.
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1 Introduction

The following standard cells are to be designed in complementary static CMOS logic in a single stage. The number of standard cells to be designed is same as the number of designers in your team. A two member team needs to design both the gates. If you are doing the project alone, you can choose one of these gates.

- NAND4 ($F = \overline{A.B.C.D}$)
- NOR4 ($F = \overline{A + B + C + D}$)
- AOI22 ($F = \overline{(A_1.A_2) + (B_1.B_2)}$)

(To understand this naming convention of the AOI cell you can refer to https://www.youtube.com/watch?v=6bKcCrAildU&list=PLP-rjhz_nIi4kBIvyb6dN60mYSNSWJioV&index=11 from 44:00 to 47:30)

2 Requirements

The following needs to be completed for each gate that you have chosen.

1. Implement the function in a single stage using transistors at schematic level. Clearly explain how the sizes of the NMOS and PMOS transistors were chosen.
2. Simulate the propagation delay of the gate from $A \rightarrow F$ ($A_1 \rightarrow F$, if you are designing AOI22). As we know, the propagation delays are a function of both input transition time and the output capacitance. Please characterize the propagation delays for input transition time of {20 ps, 100 ps} and output capacitance of {10 fF, 50 fF}. There are various input patterns involving a toggle at A , resulting in a toggle at F . For this project, we will do a limited characterization and estimate delays for one input pattern per low-to-high and high-to-low delay estimation. We will characterize the high-to-low and low-to-high propagation delays from $A \rightarrow F$, assuming all other inputs remain constant.

For NAND4, the input sequence $ABCD : 1111 \rightarrow 0111$ can be used to measure low-to-high propagation delay. Note how B , C and D are not toggling here. Similarly, identify the input pattern one can use to estimate the high-to-low propagation delay from $A \rightarrow F$.

You may organize this information in the format shown in Table 1.

t_{pLH} from A to F ($ABCD : 1111 \rightarrow 0111$)			t_{pHL} from A to F (?)		
Load capacitance	10 fF	50 fF	Load capacitance	10 fF	50 fF
Input transition			Input transition		
20 ps			20 ps		
100 ps			100 ps		

Table 1: Propagation delay from A to F in NAND4 based on schematic simulations.

Identify such input patterns for the NOR4 and AOI22 cells. If there are more than one input pattern that can result in the output transition you require keeping B , C , and D constant, you may choose one of them.

3. Complete the layout adhering to the following guidelines. The output capacitance is not part of the layout.
 - (a) All the gates from the team should have the same height. The widths can vary across the designs.
 - (b) The VDD rail should be at the top. The width of the VDD rail should be same for all gates.
 - (c) The GND rail should be at the bottom. The width of the GND rail should be same for all gates.
4. Clear the design rule check (DRC). This ensures that your design can be fabricated by the foundry.
5. Clear the layout vs schematic (LVS) check. This ensures that your schematic and layout have the same underlying netlist (except for the interconnect parasitics).
6. Run a parasitic extraction (PEX) from the layout and generate a new netlist for the circuit including these parasitics.
7. Repeat step 2 and step 3 with the layout-extracted simulations.

3 Model files

- Please use the transistors from gppdk 180nm.
- Tutorial videos on this will be uploaded to mooKIT.

4 Tool Details

- Additional tools are required for DRC, LVS and PEX. A new virtual box image will be provided for the same.
- If there are space constraints on your laptop, you may delete the old virtualbox image and keep only the new virtualbox. **Please remember to transfer any important design files you may require from the old virtualbox to the new one.** (This may be required for other circuits courses you are doing this semester.)

5 Report format

Please include the following details in the report.

- Information about the team, standard cells designed and details on who designed each standard cell.
- Did you begin by designing a reference CMOS inverter? If so, explanation on how the reference CMOS inverter sizing was chosen.

First standard cell

- Screenshot of the schematic.
- Explanation on how the transistor sizes were chosen.
- Screenshot of the layout with the DRC log. Mark the height and width of the layout in the screenshot.
- Screenshot of the layout with the LVS log.
- Table with the propagation delay values from the schematic simulation in the format shown in Table 1.
- Table with the propagation delay values from the post-layout simulation in the format shown in Table 1.
- Waveform from transient analysis of input and output transitions of the gate for any one conditions for which you simulated the delay. Mention the conditions clearly in the figure caption. Give the waveforms from both schematic and layout-extracted simulations for both 10 fF and 50 fF fanout capacitance, for 20 ps input transition time.

Second standard cell

Repeat the same as the above.